PRE-IDEATHON TEMPLATE

GENESIS IDEATHON

TEAM & DOMAIN

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- Selected Domain – #2
- Problem Statement – When a heavy electric truck is fully charged or running hot, the system automatically shuts off regenerative braking to protect the battery. This forces the vehicle to rely entirely on mechanical brakes, causing extreme overheating, rapid pad wear, and dangerous brake fade on highways and slopes. Current fixes just use heavy resistors to dump this massive amount of power as wasted heat.

DOMAIN PROBLEM IN OUR WORDS

- ***WHAT EXACTLY NEEDS TO BE SOLVED?***

Imagine driving a fully loaded, **80,000-pound electric semi-truck** down a steep highway. Normally, electric trucks use regenerative braking, using the motor to slow down while recharging the battery. But there is a dangerous catch. If the truck starts its trip with a full battery, or if the system runs hot, the computer shuts off regenerative braking to protect the battery cells.

Suddenly, the driver must rely entirely on standard mechanical brakes to stop a massive, speeding vehicle. On long hills, these friction brakes get pushed past their limits. They quickly overheat, wear out, and suffer from **dangerous brake fade**—meaning they lose stopping power entirely. Current industry fixes just use heavy resistor grids that act as giant onboard heaters to dump this massive energy as wasted heat. We need a safe, lightweight way to handle these intense energy surges without frying the battery or destroying the brakes.

- ***WHY IS IT IMPORTANT RIGHT NOW?***

This matters immediately because the **heavy electric truck transition** is already happening. Shipping fleets are rolling out thousands of electric vehicles to meet emission goals. As these massive trucks fill up public highways and mountain passes, we cannot afford a hidden vulnerability in their primary safety systems. Relying on mechanical backup brakes on steep descents is a catastrophic accident waiting to happen.

Furthermore, burning off megawatt-level energy defeats the purpose of clean transit. We are building advanced electric vehicles, only to throw away pure, clean power as thermal waste through heavy resistors that add dead weight. Solving this right now ensures that heavy freight transport is truly efficient and safe.

WHO IS AFFECTED?

TARGET USERS, STAKEHOLDERS, BENEFICIARIES

Category	Who is getting effected	Reason
Users	• Heavy Electric Truck and Bus Drivers	• They directly operate the 80,000-pound vehicles and face life-threatening danger when regenerative braking fails and friction brakes overheat on mountain descents.
Stakeholders	• Fleet Owners & Logistics Companies • OEMs & EV Engineers	• Fleets: Invest in electric trucks for emission goals but face high maintenance costs, brake degradation, and liability. • OEMs: Responsible for vehicle safety, battery health management, and regulatory compliance.
Beneficiaries	• Truck Drivers • Fleet Operators • Highway Motorists and General Public	• Truck Drivers gain a safe, reliable, and predictable braking system that doesn't put their lives at risk when starting a route with a 100% charged battery. • They save significant money on brake replacement and maintenance, avoid costly accident liabilities, and no longer lose cargo capacity to heavy, dead-weight resistor grids. • Highway Motorists and the general public are protected from the catastrophic hazard of runaway 80,000-pound trucks careening down mountain passes.

CURRENT SITUATION & CONTEXT

Existing scenario, background and how things work today

- As the current transport industry transitions to electric trucks and buses, manufacturers face a critical hurdle i.e. **High level of brake wear** due to the heavier build if electric trucks and the safety risks that follow this process. Because heavy **EVs quickly max out or overheat** their batteries on long downhills, safety systems **automatically shut off** regenerative braking which is a major contributor to maintaining the braking systems of the vehicle at a level where they can be used safely. This forces these massive vehicles to rely entirely on **mechanical friction brakes.** The extreme weight causes **rapid thermal degradation and dangerous brake fade**.
- To fix this, currently 2025–2026 industry research focuses on **multi-tiered systems** that **blend regenerative, friction**, and **hydraulic auxiliary braking** to safely split the thermal load.

Step by step workflow:

- Electric trucks and buses rely on regenerative braking on long hill sections of the road.
- Continuous charging due to regenerative breaking causes the battery to fill up or overheat.
- The system disables regenerative braking to protect the battery.
- This forces reliance on mechanical brakes which are subject to rapid, dangerous thermal fade which may cause break failures.

KEY CHALLENGES

Main difficulties, bottlenecks, root causes and consequences

- Relying only on physical breaking in electric trucks and buses [which generally are heavier than normal trucks] is not reliable due to high rate of break fade i.e. high rate of degradation of brakes which would cause break failures. Usually, we have regenerative breaking supporting the breaking effort but if the battery is either full or overheated, the truck or the bus is unable to use regenerative breaking in order to protect the battery life.

The main problem is to find a way to utilize the additional energy generated in a useful way, which at the same time should be a long term, reliable as well as a cost-effective solution to this problem.

- Root cause of the problem is that when the battery is fully charged or overheated the battery can't accept more electrical energy from the regenerative breaks.
- If this problem is left unsolved then there would be higher risks of break failures in electric trucks and buses due to heavy dependence on physical braking systems which fade fast due to the enormous weight of the electric truck and buses.

EXISTING SOLUTIONS / APPROACHES

What is already being done, and where it falls short

1.”Resistive” Regen-Braking Systems:-

•Existing solution: Diverts excess regenerative electricity into onboard resistor banks, where it is converted into heat.
•Limitations: Adds weight and cooling requirements while wasting recoverable electrical energy as heat.
•Key challenge: How can excess braking energy be stored or reused instead of being dissipated as heat

2. *Advanced Air-Braking Systems:-*

•Existing solution: Electronic Braking Systems (EBS) coordinate regenerative braking with pneumatic friction brakes to maintain controlled deceleration.
•Limitations: When regeneration is unavailable, sustained braking still places a heavy thermal load on friction brakes.
•Key challenge: How can braking safety be maintained while reducing friction-brake overheating and wear during long downhill descents

3.*Electrically Powered Trailers with Regenerative Braking:-*

•Existing solution: Electrically powered dollies or trailers use their own motors and energy storage to provide additional propulsion and regenerative braking.
•Limitations: Adds vehicle cost, mass, mechanical complexity, and extra articulation requirements.
•Key challenge: How can additional braking-energy capacity be provided without the expense and complexity of an additional powered vehicle unit

PROBLEM INSIGHTS & OPPORTUNITY AREAS

Resources required to solve the problem

- ***Hardware Requirements***

Vehicle & Powertrain Components: Electric motors, power electronics (inverters/converters), and auxiliary energy storage/dissipation hardware (e.g., supercapacitors, advanced thermal management systems, or modular auxiliary packs).

Sensors & Control Units: High-precision battery management system (BMS) sensors, temperature sensors, inclinometers/gradient sensors, and wheel speed sensors to detect descent angles and battery state-of-charge in real-time.

Testing & Prototyping Gear: Dynamometers, thermal imaging cameras (to monitor friction brake and resistor temperatures), and hardware-in-the-loop (HIL) test benches.

- ***Software Requirements***

BMS & Vehicle Control Software: Advanced algorithms to dynamically manage energy flow, predictive descent mapping (GPS/GIS integration to anticipate mountain passes), and coordinated braking control (blending friction and electric braking).

Simulation & Modeling Tools: MATLAB/Simulink or ANSYS for vehicle dynamics, electrical load simulation, and thermal dissipation modeling.

Fleet Telematics Dashboard: Software for real-time monitoring of truck health, route energy profiling, and maintenance alerts.

- ***Other resources (Including Human Resources)***

Automotive / EV Powertrain Engineers: Specialists in battery management systems and high-voltage electrical architecture.

Control Systems Engineers: Experts in software integration and dynamic braking control algorithms.

Mechanical / Thermal Engineers: Specialists in heat dissipation, lightweight materials, and heavy-duty braking systems.

Data & Testing Facilities: Access to heavy-vehicle testing tracks, topographical gradient data sets, and regulatory compliance standards (such as SAE or ISO truck safety guidelines).

INITIAL DIRECTION & EXPECTED OUTCOME

What Is Your Expected Outcome?

- ***OUR TEAM INTENTION***

Our first idea is **magnetic braking**. By using the rotational energy of the tires to **power brake blocks packed with copper coils**, we can induce a magnetic field that **provides braking force**. As a bonus, the braking effect gradually reduces as the tires slow down, making the deceleration smoother and more progressive.
Second, we are looking at **supercapacitors**. These would store excess energy harvested from the tires, which could then be used in specific scenarios—such as **powering auxiliary systems or slowly recharging a depleting main battery.**
Our last idea is a purely mechanical alternative to these electrical and chemical solutions: a **flywheel**. This involves a **solid disk spinning at high RPMs inside a vacuum-sealed chamber**. It captures rotational energy directly from the tires and can later release it to provide additional torque when needed.

- ***WHAT WE WANT TO IMPROVE***

We want to focus on refining our first two approaches—**magnetic braking and supercapacitors**—by rigorously testing and researching on their technical feasibility, while keeping the **flywheel** approach entirely in reserve as a reliable backup.

- ***THE ACHEIVEMENT WE WANT***

Our main goal is to prove our motorsports-inspired system actually works in the real world. We want to show that magnetic braking can deliver smooth, efficient stopping power while supercapacitors successfully capture and redeploy that energy.

TEAM
Your Team Name

TEAM CONTRIBUTION

GENESIS

Who will handle which part of the execution

Team Member	Role	Responsibility in Execution
Member 1 name	Role	What they will handle
Member 2 name	Role	What they will handle
Member 3 name	Role	What they will handle
Member 4 name	Role	What they will handle